Demonstration of Predictive Analytics to Analyze Microarray Data

Progarm:

```
import pandas as pd
from sklearn.model_selection import train_test_split
sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report
df = pd.read_csv("/content/leukemia_gene_expression.csv")
df.dropna(inplace=True)  X = df.iloc[:, :-1] y = df.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
scaler = StandardScaler() X_train =
scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)  model =
SVC(kernel='linear') model.fit(X_train, y_train)
y_pred = model.predict(X_test)  print("Accuracy:",
accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

Output:

```
Accuracy: 0.33
              precision    recall  f1-score   support

         ALL       0.35      0.45      0.39        74
         AML       0.35      0.36      0.36        77
     Healthy       0.19      0.10      0.13        49

    accuracy                           0.33       200
   macro avg       0.30      0.30      0.29       200
weighted avg       0.31      0.33      0.32       200
```